

Expectation-Maximization Algorithm

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⊕ Censored Exponentially Distributed Survival Times

We suppose W is a non-negative random variable having an exponential distribution with mean μ . Thus its probability density function (P.D.F.) is given by

$$f(w; \mu) = \frac{1}{\mu} \exp\left\{-\frac{w}{\mu}\right\} I_{(0, \infty)}(w), \quad (\mu > 0), \quad (1)$$

where the indicator function

$$I_{(0, \infty)}(w) = \begin{cases} 1, & \text{for } w > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

The distribution function is given by

$$F(w; \mu) = \left\{1 - \exp\left(-\frac{w}{\mu}\right)\right\} I_{(0, \infty)}(w).$$

In survival or reliability analyses, a study to observe a random sample $W_1, \dots, W_n$ from (1) will generally be terminated in practice before all of these random variables are able to be observed. We let

$$\mathbf{y} = (\mathbf{y}_1^T, \dots, \mathbf{y}_n^T)^T$$

denote the observed data, where

$$\mathbf{y}_j = (c_j, \delta_j)^T$$

and $\delta_j = 0$ or 1 according as the observation W_j is censored or uncensored at c_j ($j = 1, \dots, n$). That is, if the observation W_j is uncensored, its realized value w_j is equal to c_j , whereas if it is censored at c_j , then w_j is some value greater than c_j ($j = 1, \dots, n$).

We suppose now that the observations have been relabelled so that $W_1, \dots, W_r$ denote the r uncensored observations and $W_{r+1}, \dots, W_n$ the $n - r$ censored observations. The likelihood function for μ formed on the basis of $\mathbf{y}$ is given by

$$\begin{aligned} L(\mu) &= \prod_{j=1}^r f(w_j; \mu) \cdot \prod_{j=r+1}^n (1 - F(c_j; \mu)) \\ &= \prod_{j=1}^r \frac{1}{\mu} \exp\left\{-\frac{w_j}{\mu}\right\} \cdot \prod_{j=r+1}^n \exp\left\{-\frac{c_j}{\mu}\right\} \\ &= \left(\frac{1}{\mu}\right)^r \cdot \exp\left\{-\sum_{j=1}^n \frac{c_j}{\mu}\right\} \end{aligned} \quad (2)$$

The **log likelihood** function for μ formed on the basis of $\mathbf{y}$ is given by

$$\ell(\mu) = \log L(\mu) = -r \log \mu - \sum_{j=1}^n \frac{c_j}{\mu}. \quad (3)$$

In this case, the MLE of μ can be derived explicitly from equating the derivative of (3) to zero to give

$$\hat{\mu} = \sum_{j=1}^n \frac{c_j}{r}.$$

Thus there is no need for the iterative computation of $\hat{\mu}$. But in this simple case, it is instructive to demonstrate how the EM algorithm would work.

The complete-data vector $\mathbf{x}$ can be declared to be

$$\begin{aligned} \mathbf{x} &= (w_1, \dots, w_n)^T \\ &= (w_1, \dots, w_r, \mathbf{z}^T)^T, \end{aligned}$$

where

$$\mathbf{z} = (w_{r+1}, \dots, w_n)^T$$

contains the unobservable realizations of the $n - r$ censored random variables. In this example, the so-called unobservable or missing vector $\mathbf{z}$ is potentially observable in a data sense, as if the experiment were continued until each item failed, then there would be no censored observations.

The **complete-data log likelihood** is given by

$$\begin{aligned} \ell_c(\mu) &= \log L_c(\mu) = \sum_{j=1}^n \log f(w_j; \mu) \\ &= -n \log \mu - \frac{1}{\mu} \sum_{j=1}^n w_j. \end{aligned}$$

As $\ell_c(\mu)$ can be seen to be linear in the unobservable data $w_{r+1}, \dots, w_n$, the calculation of $Q(\mu; \mu^{(k)})$ on the E-step (on the $(k + 1)$ th iteration) simply requires each such w_j to be replaced by its conditional expectation given the observed data $\mathbf{y}$, using the current fit $\mu^{(k)}$ for μ (Why?).

By the lack of memory of the exponential distribution, the conditional distribution of $W_j - c_j$ given that $W_j > c_j$ is still exponential with mean μ . Then, we have that

$$\begin{aligned} E_{\mu^{(k)}}(W_j \mid \mathbf{y}) &= E_{\mu^{(k)}}(W_j \mid W_j > c_j) \\ &= c_j + E_{\mu^{(k)}}(W_j) \\ &= c_j + \mu^{(k)} \end{aligned} \quad (4)$$

for $j = r + 1, \dots, n$.

On using (4) to take the current conditional expectation of the complete-data log likelihood $\ell_c(\mu)$, we have that

$$Q(\mu; \mu^{(k)}) = -n \log \mu - \frac{1}{\mu} \left\{ \sum_{j=1}^r c_j + \sum_{j=r+1}^n (c_j + \mu^{(k)}) \right\}$$

$$= -n \log \mu - \frac{1}{\mu} \left\{ \sum_{j=1}^n c_j + (n-r)\mu^{(k)} \right\}. \quad (5)$$

Concerning the M-step on the $(k+1)$ th iteration, it follows from (5) that the value of μ that maximizes $Q(\mu; \mu^{(k)})$ is given by the MLE of μ that would be formed from the complete data, but with each unobservable w_j replaced by its current conditional expectation given by (4). Accordingly,

$$\begin{aligned} \mu^{(k+1)} &= \left\{ \sum_{j=1}^r c_j + \sum_{j=r+1}^n E_{\mu^{(k)}}(W_j \mid \mathbf{y}) \right\} / n \\ &= \left\{ \sum_{j=1}^r c_j + \sum_{j=r+1}^n (c_j + \mu^{(k)}) \right\} / n \\ &= \left\{ \sum_{j=1}^n c_j + (n-r)\mu^{(k)} \right\} / n. \end{aligned}$$